



CFD simulation of multiphase flows and erosion predictions under annular flow and low liquid loading conditions



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ABSTRACT

For many industrial applications dealing with gas-liquid flow, transporting gas and liquid simultaneously causes many challenges. Prediction of erosion in multiphase flow is a complex problem due to lack of accurate models for calculating particle impact velocities in multiphase flow. Computational Fluid Dynamics (CFD) studies of annular flow are being conducted to investigate annular flow behavior and particle impact characteristics. Volume of Fluid (VOF) method was employed for air-water flow simulation with high gas velocities and low liquid rates. Simulation results are compared with results obtained with Eulerian-Eulerian with Multi-Fluid VOF approach with the same flow rates. CFD simulations of annular flow were also compared and validated with experimental data previously obtained with wire-mesh sensors. The liquid film thickness in bends can significantly affect erosion, therefore the simulation results of liquid film thickness trends are also investigated based on the gas and liquid flow rates.

Particle tracking along with flow simulations have been solved simultaneously, and the maximum values of mass removal calculated with current erosion models were recorded in order to calculate the maximum erosion magnitudes. Particle impact characteristics including particle impact velocities, particle impact angles as well as particle impact number distribution along the bend were studied in several cases with different superficial gas and liquid velocities. Predicted erosion ratios are compared with available experimental data and show very good agreement.

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1. Introduction

Pipe fittings such as elbows and tees are often used in oil and gas production systems and are vulnerable to erosion damage when sand production is anticipated. Annular flow and low liquid loading flows are mostly observed in multiphase gas production pipelines and these flow patterns are characterized by the presence of high gas flow at the center of the pipe and liquid film around the pipe wall. Three major characteristics of annular flow can be stated as follows: droplet entrainment into the gas core, deposition of a fraction of entrained droplets onto the gas-liquid interface and existence of waves at the interface. When the difference between gas and liquid velocities is high enough, large shear velocities are induced due to high gas velocity that result in high interfacial shear stress and consequently a flow rate of liquid droplets is generated at the interface of two phases into the gas core. When sand particles are entrained, the same mechanism that entrains liquid can cause sand entrainment in the gas core. Thus,

sand particles that are entrained in the gas core are moving with the high gas velocity in the core and impact pipe fittings such as elbows and tees [1]. Therefore, Study of erosion modeling and prediction are important and challenging for oil and gas companies. Recent studies have shown a good progress in erosion modeling and CFD erosion analysis. McLaury et al. [2] presented an approach for predicting erosion in the vertical annular flow pattern. In the model, the erosion is assumed to result from both particles in the core and in the liquid film and summed to determine the total erosion value. Peng et al. [3] showed CFD capability in erosion prediction in multiphase flow under steady state conditions and compared with prediction models. Barton et al. [4] showed that CFD can be used to predict experimental erosion measurements to a high degree of accuracy and the best erosion prediction method under multiphase flow condition is explicitly model gas, liquid and sand as separate phases. Parsi et al. [5] also used CFD to calculate erosion rates in slug/churn flows using a multi-fluid VOF model.

There are basically three main approaches for CFD simulation of annular flow; the most dominant method is two-fluid method, second is three-fluid model and third is Eulerian-Lagrangian method. Liu et al. [6] proposed a new CFD model for vertical

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upward annular flow. In this method, mass and momentum equations are solved for the gas core region as well as liquid film with Fluent software, and the mass transfer effect with liquid droplets due to entrainment and deposition processes is accounted by a source term in the mass conservation equation and another additional source term is taken into account in order to consider velocity acceleration of entrained droplets into the gas core. A three-fluid model was introduced by Alipchenkov et al. [7] where dispersed-annular flow was considered as gas, droplets and film which is based on one-dimensional equations of mass, momentum and energy for each of the three fluids. A transport equation was considered for the variation of the number density of droplets due to deposition, entrainment, coagulation and breakup. Adechy and Issa [8] introduced an Eulerian-Lagrangian approach which is based on transport equations for the dispersed droplets and gas phase in the core and simplified equations for the liquid film, using CFD. A Lagrangian approach was employed to predict the trajectories of droplets through the gas core. The methodology accounts for the interaction between the phases including mass exchange due to entrainment and deposition.

Several authors have also simulated annular flow in pipes and two-dimensional (2D) geometries; for example, Kishore and Jayanti [9] developed a finite volume method-based CFD model to simulate two-dimensional annular gas-liquid flow in a duct. A form of modified wall function was used in order to consider the effect of liquid film on the gas phase. The traditional no-slip condition at the walls was replaced by a rough wall boundary condition, where the equivalent sand grain roughness is determined from both the gas flow and liquid flow conditions. The modified wall function incorporates the relationship that exists among wall shear stress, film flow rate and film thickness in annular flow. An additional scalar transport equation for the mass fraction of droplets was used and entrainment and deposition of droplets were considered as source terms and boundary conditions. Their simulation results were in good agreement with the literature data of annular flow parameters under equilibrium and non-equilibrium. Han and Gabriel [10] conducted a CFD study on simulation of the dynamic wave formation process in a two-dimensional small diameter tube with an inner diameter of 9.525 mm. Annular flow was simulated for vertical upward air-water flow and two types of wave evolution were identified. First, once an individual wave forms, it continues to deform and develop until its crest is finally sheared off into the gas core. The other phenomenon shows that the upstream wave is absorbed by the downstream one, and it provides material for further development of the downstream wave and eventually, the wave crest of the downstream wave is sheared off by the incoming gas phase. Erkan et al. [11] analyzed the droplet deposition onto a stagnant thin liquid film by coupling the volume of fluid (VOF) method in open-source CFD code, OpenFOAM, with moving particle semi-implicit method. The moving particle semi-implicit (MPS) numerical technique was developed by Xie et al. [12]. They investigated droplet collision and splashing on the moving liquid film.

A considerable amount of research has been devoted to the study of the two-phase annular flow regime. However, there remains much to be investigated and understood about that flow pattern. In particular, deeper investigation is needed to attain a thorough understanding of the internal structure of liquid film flow in elbows. There have been a number of studies of vertical and horizontal two-phase flow in straight pipes, but CFD studies of annular film thickness in elbows and curved pipes are limited. In two-phase flow, an elbow geometry causes inhomogeneous phase distribution, secondary flow, vortices, flow reversal, and film inversion. These phenomena can cause problems such as a dry-out region, erosion-corrosion, and tube failure in industrial applications, and also may cause safety issues, costly replacement and

maintenance. In this study, research is carried out to investigate vertical two-phase flow in a 76.2 mm I.D. pipe at high gas and low liquid flow rates so that the flow regime is annular flow. CFD simulations of two-phase flow modeling using commercial CFD software Fluent 14.5 are assessed and compared with experimental data. Then, erosion predictions were performed and compared with available experimental data after flow solution validation was obtained. Erosion predictions were also compared with a mechanistic model which has been developed for predicting erosion in elbows in multiphase flow. For better understanding of erosion mechanism, particle impact characteristics including impact speed and impact angle distributions were reported and investigated.

2. Numerical modeling

Volume of Fluid (VOF) model is a surface-tracking method which is designed for two or more immiscible fluids. This method is widely used in the literature in recent years [13–15], because it has less computational complexity, higher precision, and is easier to realize in comparison to other multiphase simulation methods. In VOF method, the momentum equation is shared by fluids, and a single momentum equation is solved throughout the domain. The momentum equation is dependent on the volume fractions of all phases through the density and viscosity, ρ and μ [16];

$$\frac{\partial}{\partial t}(\rho \vec{v}) + \nabla \cdot (\rho \vec{v} \vec{v}) = -\nabla p + \nabla \cdot \left[\mu (\nabla \vec{v} + \nabla \vec{v}^T) \right] + \rho \vec{g} + \vec{F} \quad (1)$$

where $\vec{F}$ in the right hand side of Eq. (1) describes the interaction forces acting on phases such as surface tension.

Volume fraction of each phase is obtained by the solution of a continuity equation for the volume fraction of secondary phase;

$$\frac{1}{\rho_q} \left[\frac{\partial}{\partial t}(\alpha_q \rho_q) + \nabla \cdot (\alpha_q \rho_q \vec{v}_q) \right] = S_{\alpha_q} + \sum_{p=1}^n (\dot{m}_{pq} - \dot{m}_{qp}) \quad (2)$$

In Eq. (2), S_{α_q} is the source term, $\dot{m}_{pq}$ is the mass transfer from phase p to phase q and $\dot{m}_{qp}$ is the mass transfer from phase q to phase p. The default value of the source term is zero, and assuming no mass transfer between the two phases, the continuity equation for the liquid phase can be written as follows;

$$\frac{\partial}{\partial t}(\alpha_q \rho_q) + \nabla \cdot (\alpha_q \rho_q \vec{v}_q) = 0 \quad (3)$$

This equation can be solved through implicit or explicit time discretization. The volume fraction of the primary-phase is computed based on the following equation:

$$\sum_{q=1}^n \alpha_q = 1 \quad (4)$$

In this study, an implicit scheme is selected for time discretization of volume fraction continuity equation, and a compressive scheme is selected for geometric reconstruction for the cells that lie near the interface between phases. The mixed fluid properties of density and viscosity are weighted by the volume fractions of the two fluids:

$$\begin{aligned} \rho &= \alpha_2 \rho_2 + (1 - \alpha_2) \rho_1 \\ \mu &= \alpha_2 \mu_2 + (1 - \alpha_2) \mu_1 \end{aligned} \quad (5)$$

The effects of surface tension along the interface can also be included in the VOF model. A constant value of 0.072 N/m is specified for surface tension coefficient. Surface tension acts to balance the radially inward intermolecular attractive force with the radially outward pressure gradient force across the surface

[16]. Surface tension in the interface is introduced in the model by the Continuum Surface Stress (CSS) model which is represented as an anisotropic variant of modeling capillary forces based on surface stresses.

The multi-fluid VOF model couples the VOF and Eulerian multiphase models. This allows tracking both sharp and dispersed interfaces while still being able to offer different flow fields for each phase. Similar to the VOF model, the volume fraction equation may be solved either through implicit or explicit time discretization. The biggest difference is that the Eulerian-Eulerian with Multi-Fluid VOF model also solves conservation equations for each phase. The conservation equations solved are given below:

The continuity equation for phase q is

$$\frac{\partial}{\partial t}(\alpha_q \rho_q) + \nabla \cdot (\alpha_q \rho_q \vec{v}_q) = 0 \quad (6)$$

The momentum equation for phase q is

$$\frac{\partial}{\partial t}(\alpha_q \rho_q \vec{v}_q) + \nabla \cdot (\alpha_q \rho_q \vec{v}_q \vec{v}_q) = -\alpha_q \nabla p + \nabla \cdot \vec{\tau}_q + \alpha_q \rho_q \vec{g} + \sum_{p=1}^n \vec{R}_{pq} \quad (7)$$

where, α_q is the volume fraction of phase q , ρ_q is the density of phase q , $\vec{v}_q$ is the velocity of phase q , p is the pressure shared by all phases, $\vec{\tau}_q$ is the q th phase stress-strain tensor, and $\vec{R}_{pq}$ is an interaction force between phases. Thus, the benefit of applying this model is that velocity information for each phase can be extracted from solution and studied for erosion modeling. For the current study, both models are used to characterize the flow field.

2.1. Flow solution validation

To validate CFD annular flow simulation and erosion prediction capability at high gas and low liquid rates, flow field data collected by wire mesh sensor from a vertical section of a standard 90° elbow is compared with simulation results from both approaches. The geometry includes a standard 90-degree elbow with a length of 10d upstream of the elbow and 2d downstream of the elbow. Pipe diameter and elbow radius of curvature were 76.2 mm and 1.5, respectively (Fig. 1).

A very fine structured mesh was generated all through the pipe and elbow where the total number of cells were about 4 million for VOF approach while a relatively coarse mesh (about 500,000 cells) was used for the Eulerian-Eulerian with VOF approach. Different meshes were compared and the mesh with about 4 million

cells was selected for the VOF method, since the simulations indicated no significant differences between this mesh and a finer mesh. A very fine mesh was created near the wall where the cell immediately adjacent to the wall is specified to have a thickness of 0.2 mm for VOF method and 0.5 mm for multi-fluid VOF method as shown in Fig. 2. A velocity-inlet boundary type was selected for the flow inlet and gas and liquid were introduced to the domain separately. Two phases were separately injected at the inlet, gas was introduced from the center of the inlet while liquid was injected circumferentially as shown in Fig. 2 with red and blue colors representing the gas phase and liquid phase, respectively. An initial value of 1 mm was specified for the liquid film thickness at the inlet. In this study, the RSM turbulence model with Scalable Wall Function was used in VOF method. RSM is a more accurate turbulence model when accounting for the effects of streamline curvature, swirl, and rotation, and it has greater potential to give accurate predictions for complex flows [16]. Time step was set as $1e-4$ s with a maximum of 30 iterations per time step. The criteria for residuals were defined as $1e-3$ for continuity while residuals of turbulent component values reach less than $1e-4$. It is worth mentioning that in multi-fluid VOF approach $k\omega - sst$ turbulent model was used which is a combination of the k -epsilon in the free stream and the k -omega models near the wall. It does not use wall functions and tends to be accurate when solving the flow near the wall [17]. Considering the case that $y^+ \sim 15$ which approximately lies in the buffer region.

In previous Erosion/Corrosion Research Center (E/CRC) experimental investigations [18,19], a wire-mesh technique was applied in order to study air-water flow behavior in a vertical 3-inch pipe. Instantaneous void fractions were calculated from the measured conductance which are attained from wire mesh sensor (WMS) experiments. Fig. 3 indicates a comparison of void fraction time series of WMS and CFD-VOF predictions at the location 10D from the inlet and before bend with superficial gas velocity of $V_{sg} = 27$ and superficial liquid velocity of $V_{sl} = 0.04$ m/s simulated with VOF method. In this work, superficial velocities are presented as

$$V_{sg} = \frac{q_G}{A_p} \quad V_{sl} = \frac{q_L}{A_p} \quad (8)$$

where A_p is the cross-sectional area of the pipe and q_G , q_L are gas and liquid volumetric flow rates, respectively. It should be mentioned that presented values in Fig. 3 are cross-sectional averaged void fraction data and WMS data was acquired at a frequency of 10,000 Hz for experiment time of 60 s and calculated averaged void fractions from CFD are reported for every time step of simulation ($1e-4$ s). Void fraction fluctuations are due to the presence of waves and cyclic liquid structures. From Fig. 3, it is seen that regardless of fluctuations there is good agreement in void fraction average magnitude between experimental data and CFD results. These discrepancies might be resolved by increasing the number of cells in the simulation domain as well as decreasing simulation time steps similar to experimental data acquisition frequency. Although, this will increase the computational costs but will help to simulate multiphase flow more accurately. Liquid film thickness play an important role in CFD erosion prediction in multiphase flow and in these simulations, average void fractions which represent liquid film thickness are barely distinguishable from those of the experimental data.

A histogram plot of void fraction distribution predicted with CFD and measured void fractions are presented in Fig. 4. It can be concluded that there is a small discrepancy between the void fractions that show the maximum value of probability in experiment and CFD. The maximum value of probability in the CFD result shows a void fraction of 0.95, while for WMS data, this value is 0.96. The figure also indicates that the void fraction deviation is

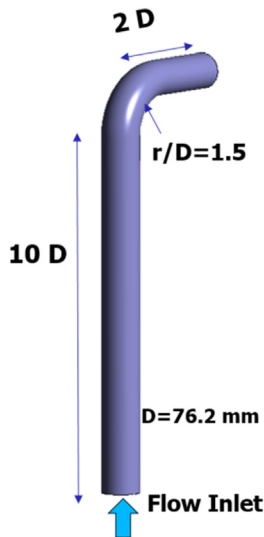


Fig. 1. Geometry for CFD simulations of two-phase flow in an elbow.

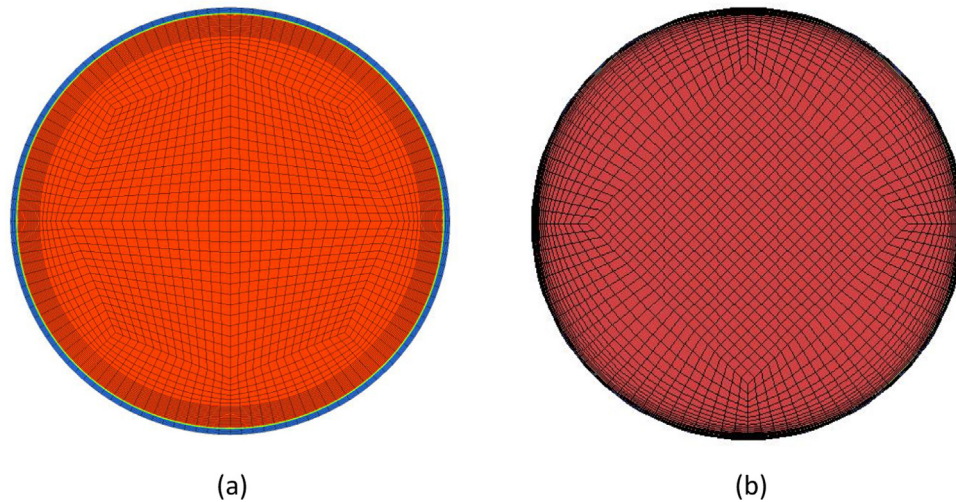


Fig. 2. Mesh for elbow geometry for simulations with (a) VOF method (b) multi-fluid VOF method.

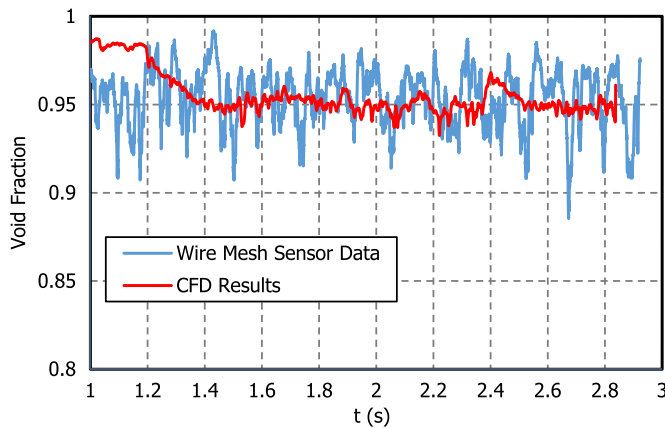


Fig. 3. Comparison of void fraction times series from WMS and CFD-VOF predictions at the location before bend with $V_{sg}=27$ and $V_{sl}=0.04$ m/s.

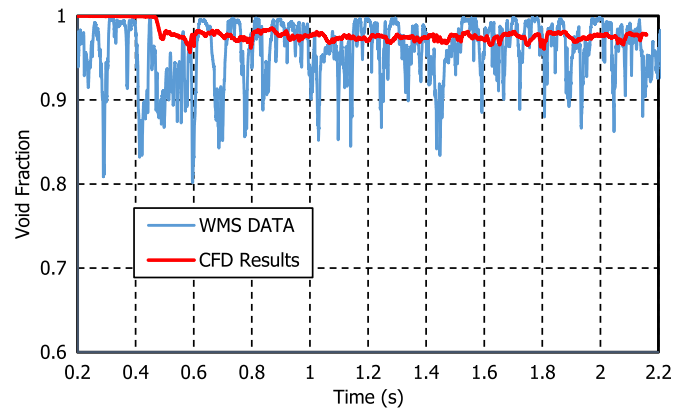


Fig. 5. Comparison of void fraction times series from WMS and CFD-VOF predictions at the location before bend with $V_{sg}=41$ and $V_{sl}=0.04$ m/s.

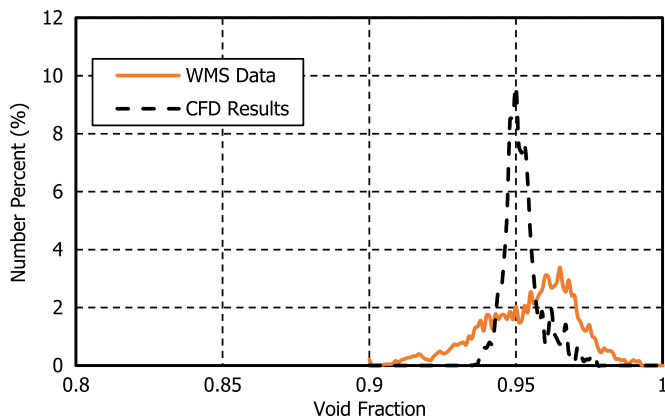


Fig. 4. Histogram plot of predicted with CFD and measured void fractions at the location before bend with $V_{sg}=27$ and $V_{sl}=0.04$ m/s.

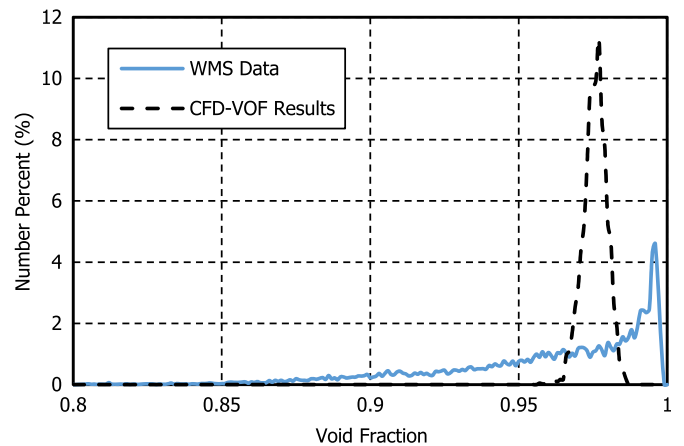


Fig. 6. Histogram plot of predicted with CFD and measured void fractions at the location before bend with $V_{sg}=41$ and $V_{sl}=0.04$ m/s.

higher in the experiment than predicted with CFD. Similar plots are presented in Figs. 5 and 6 for superficial liquid velocity of 0.04 m/s and higher gas superficial velocity of 41 m/s. Again it can be seen that the maximum probability value of void fraction obtained using CFD is lower than experiment with values of 0.97 and 0.99. The deviation in void fraction results for this condition are also larger for experiment than CFD. It is important to note that the CFD time steps are larger than the time steps between data

collection, so this may be a source of the difference in the deviation of the results.

Comparison of void fraction distributions calculated in a plane before the bend with VOF method and multi-fluid VOF method is presented in Fig. 7. In VOF method, there is one set of equations being solved during a simulation, mixture properties including velocity present in the simulation; however, in a multi-fluid VOF simulation, two phases are solved through two sets of equations

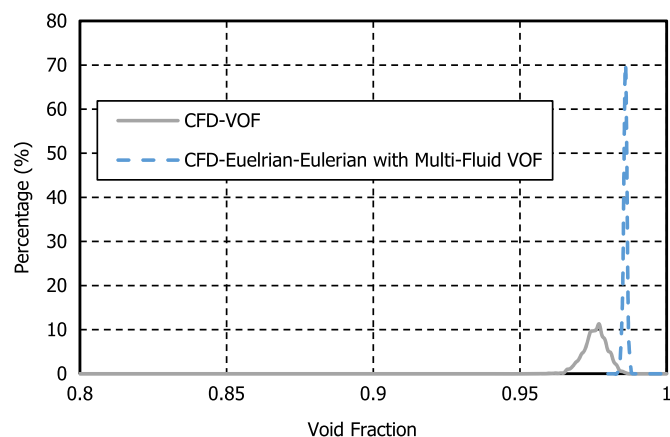


Fig. 7. Comparison of void fraction distribution calculated with VOF method and Eulerian-Eulerian multi-fluid VOF method ($V_{SG}=41$ m/s and $V_{SL}=0.04$ m/s).

and momentum and continuity equations are solved for each phase. The accuracies of these predictions are not well understood, but more accurate flow modeling is expected with the multi-fluid VOF method since in this method phases are being solved separately and interact with drag force. Simulation results of the multi-fluid VOF method indicate less fluctuation and almost steady-state flow conditions, while the simulated flow field obtained with the VOF method captured liquid film waves and fluctuations. This might be due to the better mesh quality which was used in this simulation as compared with the multi-fluid VOF simulation. It is seen that the value that shows the maximum probability of void fraction, in both methods, is comparable with experiment. It is worth mentioning that since momentum and continuity equations are solved for each phase in multi-fluid VOF method, it is much more computationally expensive even on super-computers.

Therefore, particle tracking and erosion predictions in this paper are conducted in the simulations which are done with the VOF method.

3. Results and discussion

The liquid film thickness in the bend can significantly affect erosion as particles that are entrained in the gas core have to penetrate through the liquid film to impact the pipe wall. In order to accurately predict erosion magnitude within pipe bends, prediction of liquid film thickness in bends is important. Experiments have shown that average liquid film thickness along the outer bend of the elbow is lower than that for upward vertical flow in straight sections of pipe [20]. Figs. 8 and 9 display contours of air volume fraction on an elbow simulated with Eulerian – Eulerian multi-fluid VOF method and VOF method. A thicker liquid film was observed at the inner bend in comparison with the outer bend of the elbow in both methods. Liquid distribution on the elbow extrados changes by wave movements and generation of vortices in the bend as well as gravity effects interact with the liquid film and move it toward the inner wall of the bend, and this causes a thin liquid film or dry-out region on the outer wall of the bend. The average value of liquid film thickness on the outer bend of the elbow obtained with the VOF method is 6.9×10^{-2} mm for $V_{SG}=41$ and $V_{SL}=0.04$ m/s which is comparable with 9.7×10^{-2} mm obtained from the multi-fluid VOF simulation with the same velocities.

Fig. 10 shows cross-sectional views of phase distribution at two different cross-section planes in a pipe simulated with $V_{SG}=27$ m/s and $V_{SL}=0.04$ with the VOF method. Fig. 10-a displays the contour of air volume fraction in a plane at a distance of 1D upstream of the bend inlet in the straight section of the pipe, and Fig. 10-b shows the phase distribution in the plane at 45° in the elbow. It can be seen that in the straight section, the liquid film is

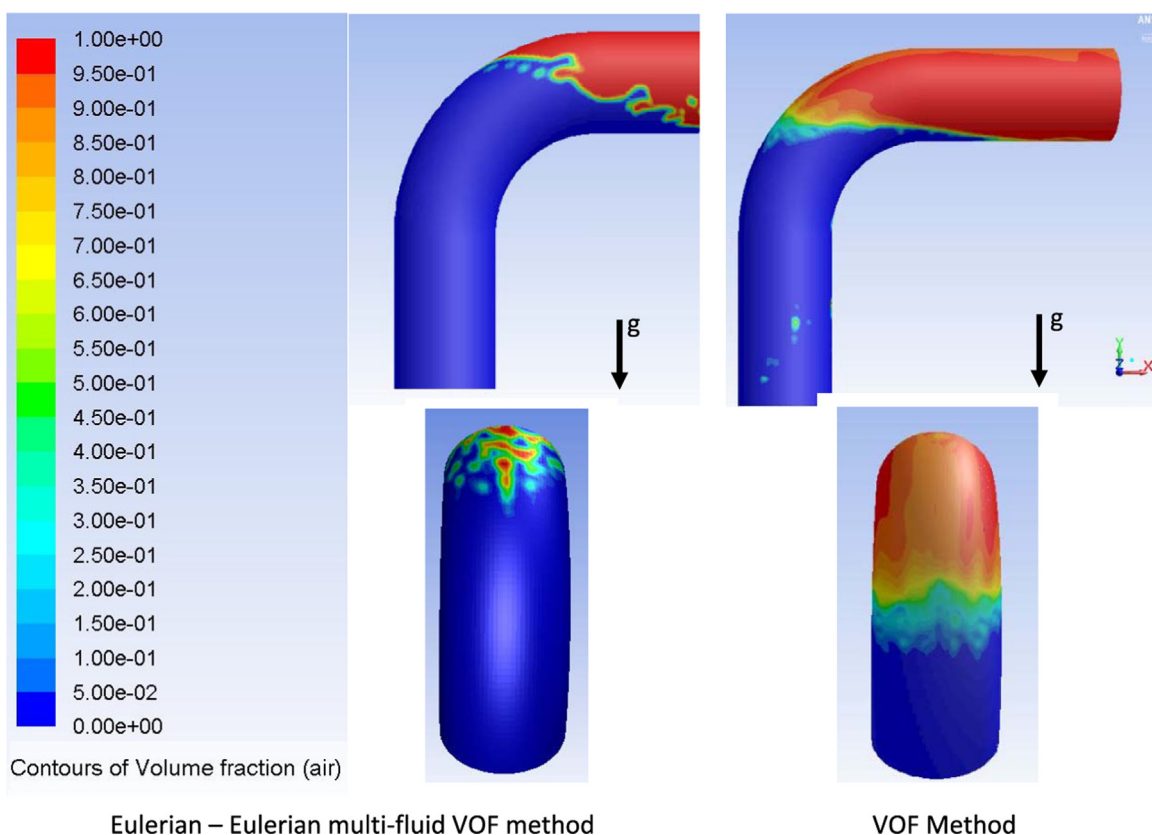


Fig. 8. Liquid phase distribution on bend for $V_{SG}=27$ m/s and $V_{SL}=0.04$ m/s.

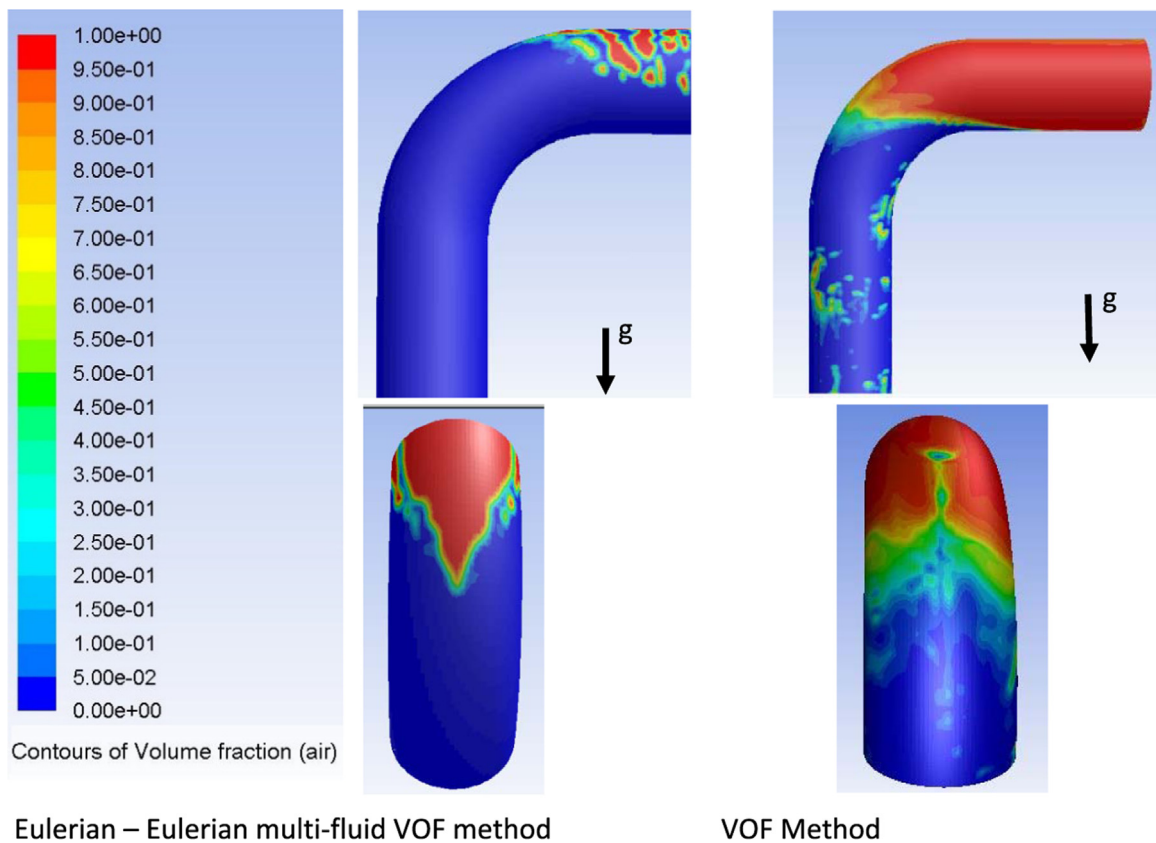


Fig. 9. Liquid phase distribution for $V_{sg}=41$ m/s and $V_{sl}=0.04$ m/s.

distributed uniformly, while at the 45-degree plane, the liquid film is thicker at the inner radius of the bend as compared with the outer radius. Fig. 11 presents tangential velocity vectors at the 45° plane, which shows the form of two concentric vortices. Some recent CFD studies of flow in elbows have shown that Dean vortices normally form when fluids flow through an elbow along with drainage of the liquid film due to gravity as well as other phenomena.

Reduction of the film thickness in elbows reduces the cushioning effect of the film on particle velocities and causes a more severe

erosion problem. Therefore, it is important to accurately predict the circumferential distribution of film thickness in elbows. Effect of superficial gas velocity on liquid film thickness is presented in Fig. 12 for superficial liquid velocity of 0.1 m/s. Liquid film thickness along the inner radius, outer radius and straight section are shown in this plot. It is shown that as gas velocity increases the average liquid film thickness along the outer radius of the elbow decreases, but there is not a significant difference seen in average film thickness with constant V_{sl} for the inner radius and straight section. However, these values change as liquid rate changes. Fig. 13

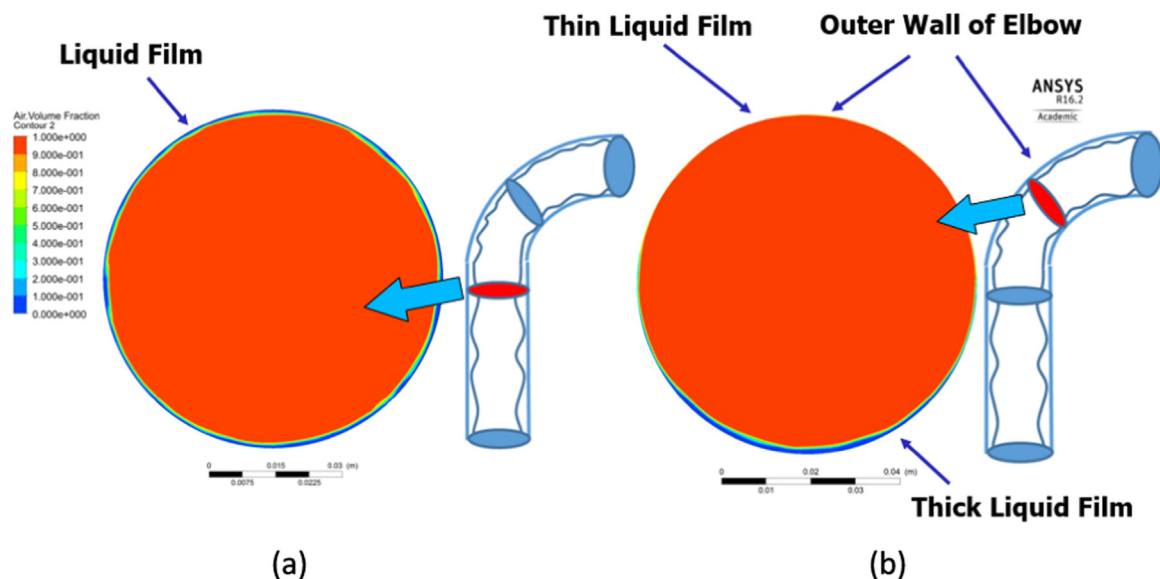


Fig. 10. Phase distribution in a) straight section of pipe b) at 45° in bend.

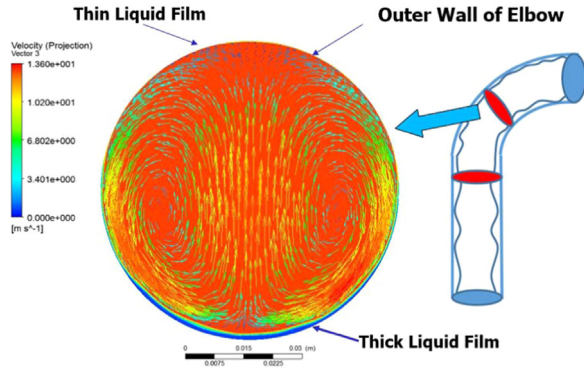


Fig. 11. Tangential velocity vectors at 45° plane for simulation with $V_{sg}=27$ m/s, $V_{sl}=0.04$ m/s.

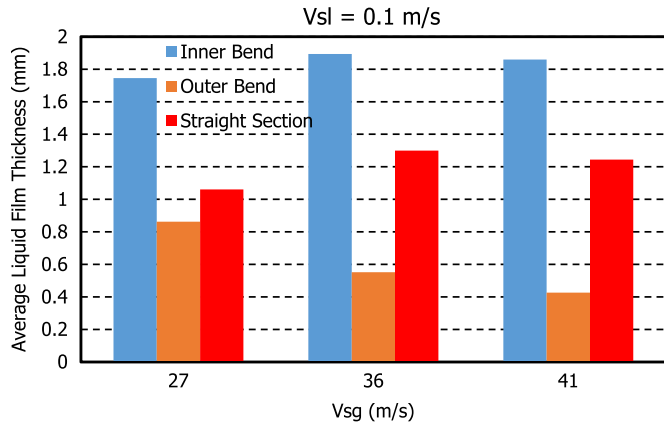


Fig. 12. Effect of superficial gas velocity on liquid film thickness with superficial liquid velocity of 0.1 m/s.

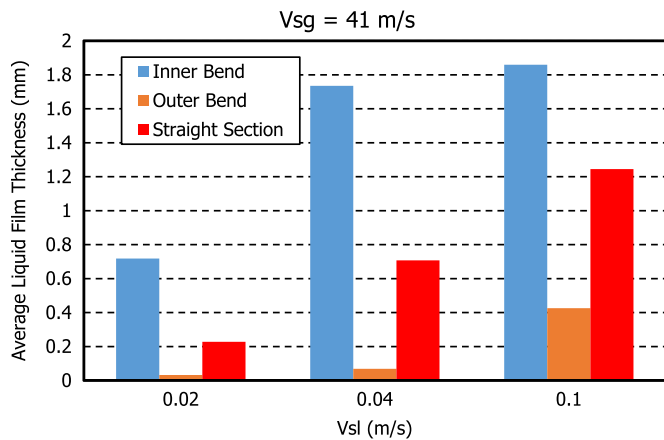


Fig. 13. Effect of superficial liquid velocity on liquid film thickness with constant superficial gas velocity of 41 m/s.

indicates that with a constant gas velocity of 41 m/s, liquid film thickness increases for all three sections as liquid rate increases. A similar trend is also observed for a lower gas velocity of 27 m/s which is shown in Fig. 14.

Ju [21] developed a new liquid film thickness model for vertical annular flow in pipes based on three databases. The model includes pressure, liquid and gas velocities, diameter and viscosity effects on liquid film thickness. He obtained the following equation which is a function of gas and liquid Weber numbers,

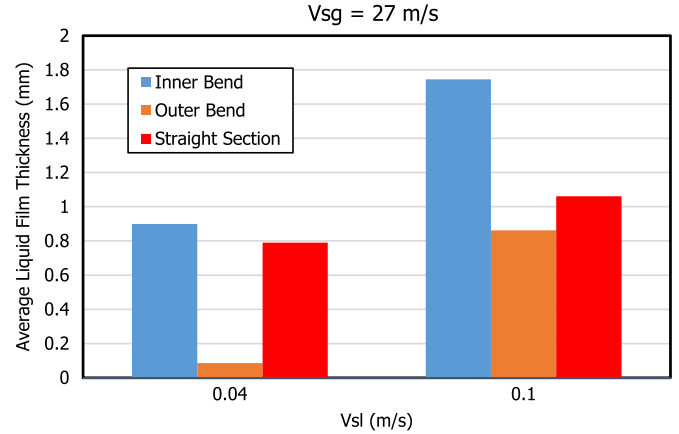


Fig. 14. Effect of superficial liquid velocity on liquid film thickness with constant superficial gas velocity of 27 m/s.

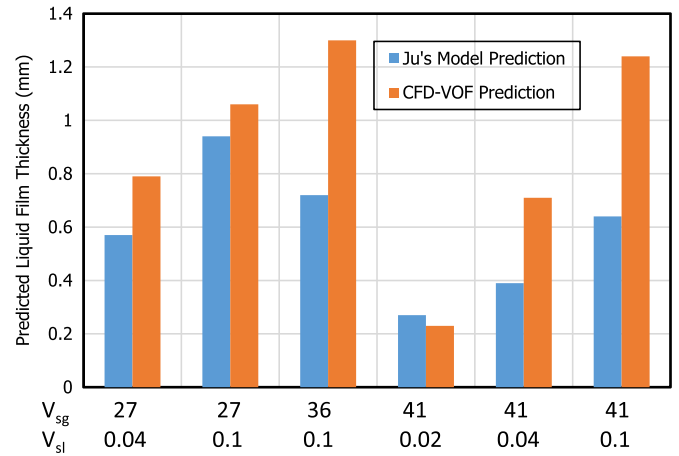


Fig. 15. Comparison of CFD simulation of liquid film thickness with model prediction.

$$\frac{\delta}{\delta_{max}} = \tanh(14.22 We_f^{0.24} We_g^{-0.47} N_{\mu_f}^{0.21}) \quad (9)$$

where We_f is liquid Weber number and We_g is gas Weber number proposed with the following expressions,

$$We_f = \frac{\rho_f j_f^2 D}{\sigma} \quad (10)$$

$$We_g = \frac{\rho_g j_g^2 D}{\sigma} \left(\frac{\Delta \rho}{\rho_g} \right)^{1/4} \quad (11)$$

where j_f , j_g are superficial liquid and gas velocities. N_{μ_f} is a non-dimensional viscosity number and is defined as,

$$N_{\mu_f} = \frac{\mu_f}{\sqrt{\rho_f \sigma} \sqrt{\frac{\sigma}{g \Delta \rho}}} \quad (12)$$

where μ_f is the liquid viscosity and g is the gravitational acceleration. Liquid film thickness prediction from CFD simulations are compared with predicted values obtained from the model proposed by Ju [21] in Fig. 15. It can be seen that there is reasonable agreement between liquid film thickness simulation in the straight section of the pipe and liquid film thickness calculated with Ju's model.

3.1. Particle tracking and erosion prediction

CFD-based erosion prediction is a three-step process: continuous carrier fluid flow modeling, particle tracking using a Lagrangian approach, and erosion calculation using the information from particle impingements on the wall. The first step is to solve the carrier fluid flow equations. Previous studies have shown the capabilities of CFD simulation in erosion prediction for single-phase flow [22]. Experimental analysis of erosion in multiphase flow is a complex problem because it requires an advanced experimental apparatus in order to measure thickness losses. Therefore, CFD could be an alternative method for erosion determination to investigate key parameters such as particle impact speed and angle which affect erosion.

After the two-phase flow simulation is solved for a few seconds, particles are injected from the inlet into the gas and liquid phases. The proposed governing equation of motion in a Lagrangian framework for a particle is

$$m_p \frac{d\vec{V}_p}{dt} = \sum \vec{F} \quad (13)$$

where m_p is particle mass, $\vec{V}_p$ is particle velocity vector and $\vec{F}$ represents external forces acting on the particle, which includes pressure effects and gravity in this study. In the current unsteady flow conditions, the two-phase flow simulation and particle tracking are solved simultaneously. It is assumed that the particle loading is sufficiently low (sand concentration is 1% by weight of water) so that the effect of particles on fluid and particle-particle interactions can be ignored. The erosion rate expression must be specified at the walls. Once a particle impacts the wall, impingement information and the location of each impingement point is recorded at every time step. Particle impingement information such as velocity and impact angle are used to calculate an erosion ratio. Parsi et al. [5] used CFD to investigate sand particle erosion in elbows, while the Multi-fluid VOF method was used for flow simulation. For particle tracking in multiphase flow with the multi-fluid VOF approach, ANSYS FLUENT uses the primary phase velocity and properties to calculate the drag force exerted upon the particle. They developed a User Defined Function (UDF) that applied the properties of the fluids surrounding the particles for particle drag force computation. However, in the current study the VOF method was used, where ANSYS Fluent utilizes mixture properties for solving the particle tracking equation. Mansouri [23] proposed a generalized erosion approach for predicting erosion magnitude in both gas-sand and liquid-sand erosion conditions. This model is developed based on cutting and deformation erosion ratios for stainless steel 316. His experiments have shown that deformation erosion magnitude is negligible for liquid-sand flows and proposed the following equation for this condition;

$$ER_c = \begin{cases} C_p F_s \frac{V_p^{2.41} \sin(\alpha) [2K \cos(\alpha) - \sin(\alpha)]}{2K^2 P} & \alpha \leq \tan^{-1} K \\ C_p F_s \frac{V_p^{2.41} [\cos^2(\alpha) - F(St) \sin^2(\alpha)]}{2P} & \alpha \geq \tan^{-1} K \end{cases} \quad (14)$$

$$F(St) = 0.24 St^{-0.503} \quad (15)$$

where C and K are empirical constants in the cutting erosion ratio equation. ρ , FS and V_p are the target material density, sharpness factor and particle impact velocity, respectively. $FS=1$ for sharp, $FS=0.5$ for semi-rounded and $FS=0.2$ for fully rounded sand particles. P is the plastic flow stress of the target material which is assumed to be the Vickers hardness H_v (GPa). St is the Stokes number which is the ratio of the particle response time τ_p to the

time scale of the fluid flow τ_f :

$$St = \frac{\tau_p}{\tau_f} \quad \tau_p = \frac{d_p^2 \rho_p}{18 \mu_f} \quad \tau_f = \frac{D}{U} \quad (16)$$

where d_p , ρ_p , and μ_f are the particle mean diameter, particle density and fluid viscosity, respectively.

The liquid-sand erosion equation is used in this study which is implemented in Fluent through a User-Defined Function (UDF) for calculating erosion magnitudes under annular flow conditions.

Due to the impingement of particles on the wall, there is a momentum loss of particles as defined by the coefficient of restitution, e , which is the ratio of particle velocity components after collision to the corresponding components before collision. A normal or tangential coefficient of restitution equal to 1.0 implies that the particle retains all of its normal or tangential momentum after the rebound (an elastic collision). A normal or tangential coefficient of restitution equal to 0.0 implies that the particle retains none of its normal or tangential momentum after the rebound [22]. Non-constant coefficients of restitution can be applied for wall zones. In order to accurately predict the particle trajectories an appropriate rebound model describing the particle-wall collision must be used. Based on experimental data for 2024 Aluminum, Grant and Tabakoff [24] proposed the following angle-dependent coefficients of restitution with standard deviations:

$$e_{Nor} = 0.993 - 1.76\theta + 1.56\theta^2 - 0.49\theta^3 \quad (17)$$

$$e_{Tan} = 0.998 - 1.66\theta + 2.11\theta^2 - 0.67\theta^3 \quad (18)$$

$$\sigma_{Nor} = -0.0005 + 0.62\theta - 0.535\theta^2 + 0.089\theta^3 \quad (19)$$

$$\sigma_{Tan} = 2.15\theta - 5.02\theta^2 + 4.05\theta^3 - 1.085\theta^4 \quad (20)$$

The presented rebound model was implemented to FLUENT using an UDF file. Particles impact the wall and the maximum value of mass removal of wall per unit area was monitored at every time step. Fig. 16 presents the maximum erosion magnitude in kg/m² versus simulation time for $V_{sg}=27$ and $V_{sl}=0.04$ m/s. The slope of this line is used to compute erosion magnitude in kg/m² s. It should be noted that for other simulation cases with higher gas velocities, the slope was observed to be higher while by increasing the liquid rate this slope decreases.

An erosion contour for a simulation with $V_{sg}=41$ m/s and $V_{sl}=0.04$ m/s is displayed in Fig. 17. Erosion values are converted to mm/kg and plotted in Fig. 18 for three conditions with $V_{sg}=27$ m/s

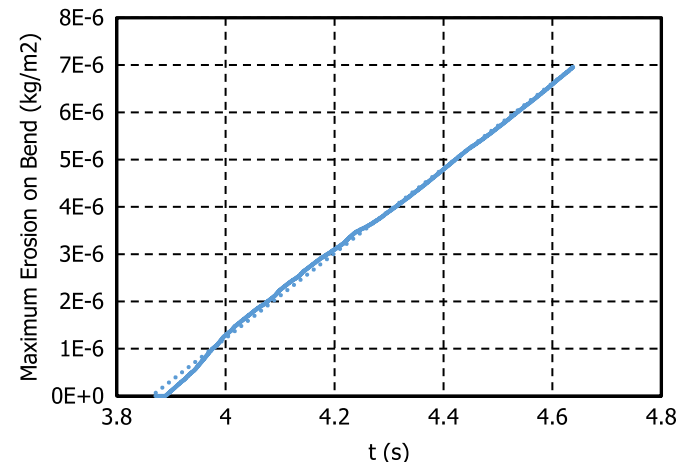


Fig. 16. Maximum erosion magnitude on bend versus simulation time for $V_{sg}=27$ and $V_{sl}=0.04$ m/s.

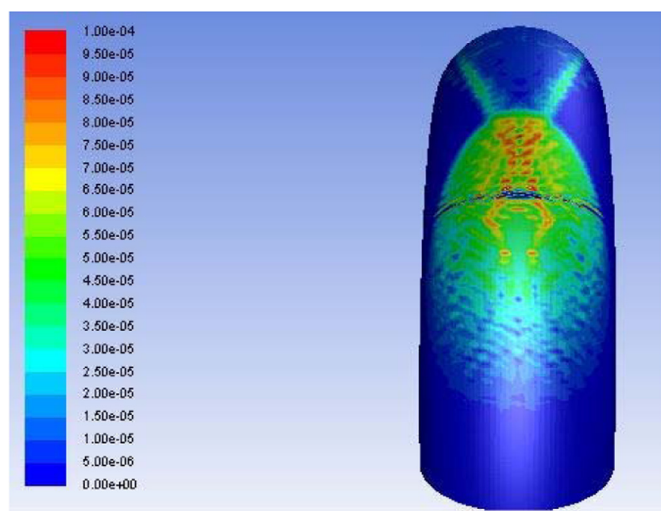


Fig. 17. Erosion Contour on elbow for simulation with $V_{sg}=41$ m/s and $V_{sl}=0.04$ m/s.

Table 1

Comparison of predicted erosion with the model developed by Mclaury et al. [2] with calculated erosion ratio in CFD simulations.

V_{sg} (m/s)	V_{sl} (m/s)	Erosion Ratio Prediction (mm/kg)	
		Model	CFD
27	0.04	4.39e-4	6.18e-4
27	0.1	4.77e-4	3.49e-4
41	0.02	1.42e-3	2.15e-3
41	0.04	1.58e-3	2.34e-3
41	0.1	1.69e-3	1.31e-3

penetrate through a thicker liquid film, consequently particle impact velocities and erosion decrease. Good agreement can be seen between experimental erosion results and CFD predictions. A similar trend is shown in Fig. 19 for a higher superficial gas velocity of 41 m/s.

Mazumder et al. [25] developed a mechanistic model for predicting erosion in annular multiphase flow considering the effects of particle velocities in gas and liquid phases of the flow. Mclaury et al. [2] developed upon the work of Mazumder and further refined the multiphase flow erosion model accounting for two-dimensional particle tracking through the annular film. This model predicts the particle impact velocity that causes erosion on the wall. Particle impact velocity is obtained through particles tracking in the gas core and in the liquid film which is influenced by liquid film thickness, film velocity, entrained liquid droplets and gas core velocity and then erosion equation is applied to predict the erosion rate using the impact information. Table 1 indicates a comparison of predicted erosion with the model developed by Mclaury et al. [2] with calculated erosion ratio in CFD simulations. It is concluded that CFD results are in consistent with predicted values with Mclaury et al. [2] model.

3.2. Particle impact characteristics distribution

Investigation of particle impact characteristics plays an important role in evaluation and modification of erosion prediction models. Erosion prediction in multiphase flows is a complex problem, since there is not an accurate model for calculating particle impact speeds and angles in multiphase flow regimes; therefore, it is important to investigate particle impact behavior. Fig. 20 shows particle impact distribution on an elbow obtained for $V_{sg}=41$ m/s at different liquid rates. It is observed that for the gas only flow condition, most particles impact with high velocities between 15 to 42 m/s; however, this range changes as liquid is added to the flow. For $V_{sl}=0.02$ m/s, most particles impact with velocities

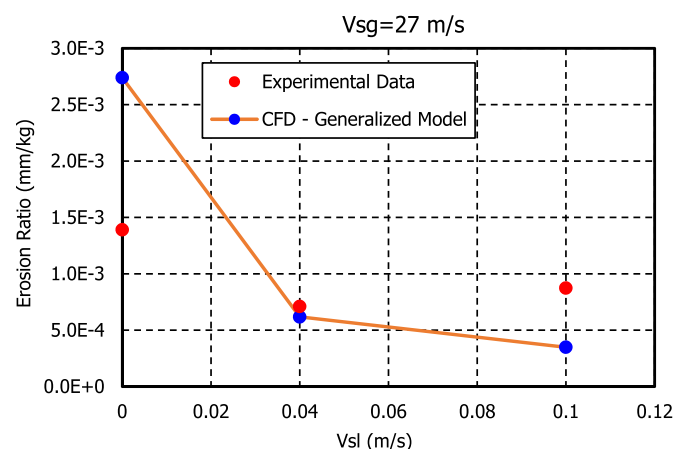


Fig. 18. Comparison of erosion experimental data with CFD predictions for $V_{sg}=27$ m/s (experimental data from [18]).

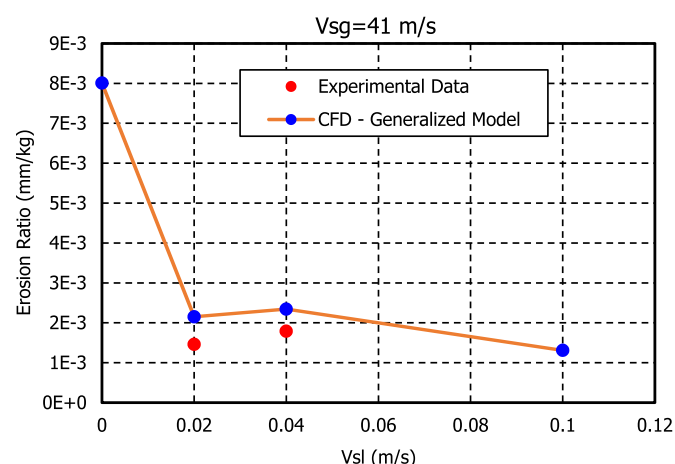


Fig. 19. Comparison of erosion experimental data with CFD predictions for $V_{sg}=41$ m/s (experimental data from [18]).

s: gas only condition and annular flow conditions with $V_{sl}=0.04$ and 0.1. Fig. 18 also contains experimental data for comparison. It can be observed that erosion magnitude decreases by increasing the liquid flow rate, since the liquid film thickness increases and particles moving with high velocities in the gas core have to

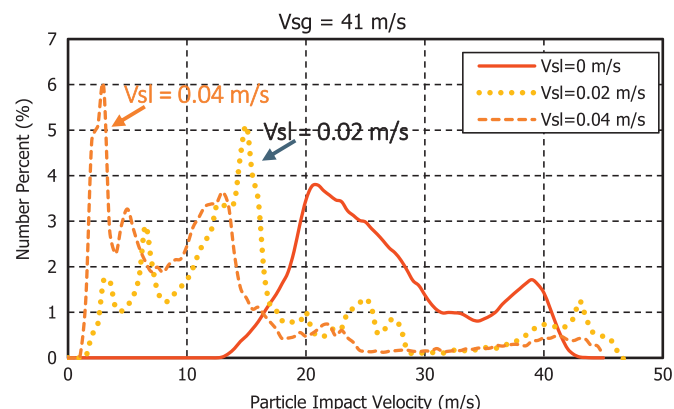


Fig. 20. Particle impact distribution on elbow obtained for $V_{sg}=41$ m/s with different liquid rates.

lower than 15 m/s. Further increasing the liquid rate to 0.04 m/s causes the particles to impact with lower velocities in comparison to the lower liquid-loading case. A similar trend is observed for a lower gas velocity of 27 m/s as shown in Fig. 21. In this figure, the number of particles impacting with high velocities decreases by increasing the liquid velocity.

Another important factor in erosion modeling, besides particle impact speed, is particle impact angle. Experiments have shown that for ductile materials, erosion increases with impact angles up to a maximum point (approximately between 15° and 30°) and then decreases to a certain value at 90° impact [26]. Fig. 22a and b show the particle impact angle distribution for $V_{sg}=41$ m/s at three different liquid rates including 0.02, 0.04 and 0.1 m/s. It is observed that most particles impact with low angles. It can also be concluded that for a constant gas velocity the majority of particles impact at lower impact angles as the liquid rate increases. This is another cause for higher erosion at lower liquid rates.

Previous experimental investigations for sand erosion in multiphase flows have shown that maximum erosion damage is mostly observed at locations between 42° and 47° along the outer radius of the bend [18] for gas dominant flows. Therefore, the following plots provide particle impact characteristics along the outer radius of the bend. The particle impact velocity distribution is shown in Fig. 23 under an annular flow condition with $V_{sg}=41$ and $V_{sl}=0.1$ m/s. It is observed that for the predicted high erosion location (between 40° and 50°), particles impact with high velocities, and these particles cause maximum erosion at bend angles

between 40° and 50°. There are some particles moving in the liquid film and impact with very low velocities which are observed at the bottom of Fig. 23, while particles moving in the gas core impact with higher velocities. Most particles impacting after 60° are rebounded particles and have lower velocities as compared with particles impacting for the first time at lower bend angles.

4. Conclusions

In this work, the capability of computational fluid dynamics for multiphase flow simulation has been tested with emphasis on erosion prediction and comparisons are made with experimental data. Comparison of multiphase flow simulation approaches play an important part in flow field investigation and developing erosion prediction models based on flow behavior. In this work, it has been demonstrated that both multi-fluid VOF method and VOF method are able to model multiphase flow under annular flow conditions with good confidence; however since the grid number and quality generated for simulations with VOF method is higher, it shows a better demonstration of disturbance waves along the pipe. The main limitations of the multi-fluid VOF method are instability and convergence problems as well as CPU time which rise exponentially with increasing number of mesh, while VOF simulations are more stable and less time-consuming since a single momentum equation is solved.

Apart from a slight discrepancy between void fraction distributions in VOF simulations with experimental data, the erosion

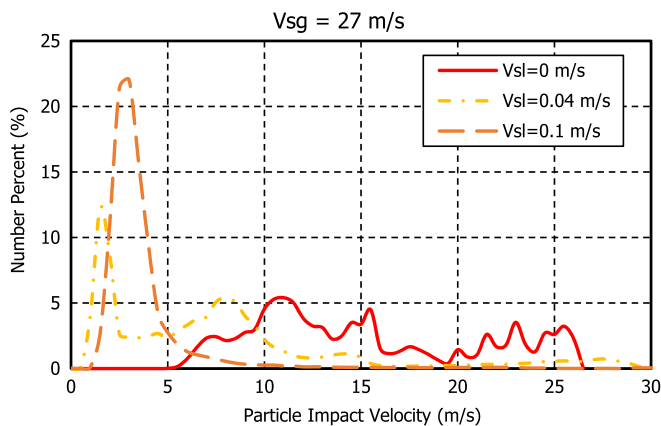


Fig. 21. Particle impact distribution on elbow obtained for $V_{sg}=27$ m/s with different liquid rates.

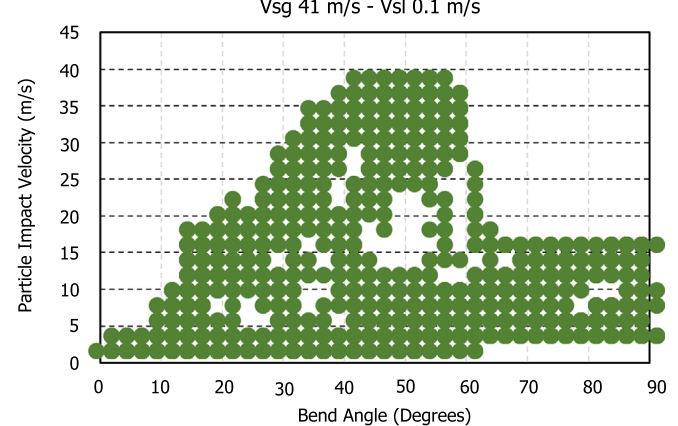
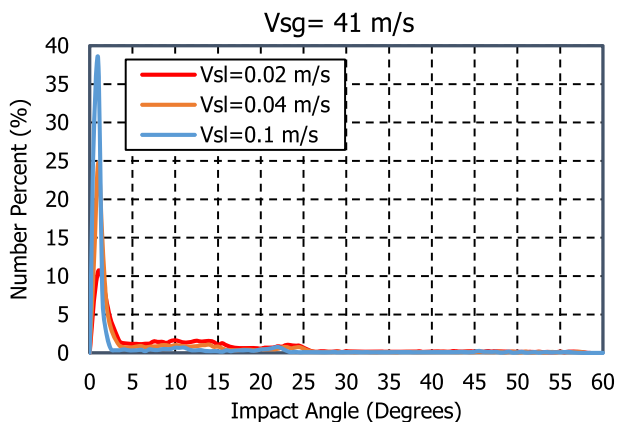
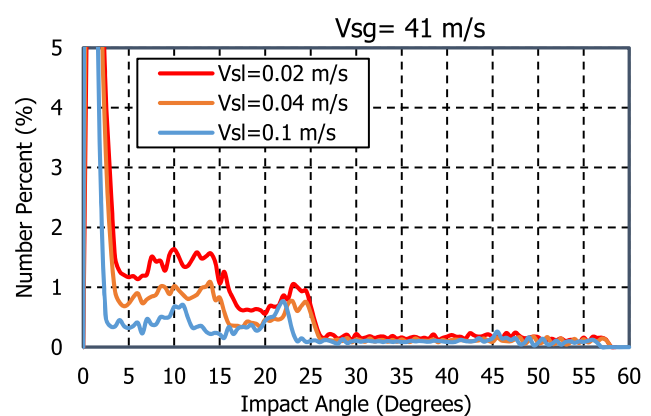


Fig. 23. Particle impact velocity distribution along the outer radius in a bend for V_{sg} 41 m/s - V_{sl} 0.1 m/s.



(a)



(b)

Fig. 22. Particle impact angle distribution for $V_{sg}=41$ m/s and $V_{sl}=0.02, 0.04, 0.1$ m/s.

prediction results indicate good agreement with previous erosion experiments. It is plausible that a number of limitations could have influenced the results obtained. First, the upstream length of the elbow may not be long enough for multiphase flow conditions to reach equilibrium conditions. Therefore, flow behavior, fluid velocities and consequently particle impact properties may change by increasing the upstream pipe length. Another possible source of error is that both investigated CFD approaches were not able to accurately simulate liquid droplet entrainment from the liquid film to the gas core with the current grid and solution methods. Liquid entrainment can influence liquid film thickness which plays an important role in erosion predictions, since particles which are traveling in the gas core have to penetrate through the liquid film to impact the wall and cause erosion damage.

A number of factors affect erosion prediction models including particle impact velocities and angles. This investigation has addressed some of these properties under annular flow and future studies on the current topic are required to explore annular flow in more detail. A larger experimental database is also necessary to study the behavior and relation between superficial gas and liquid velocities with phase velocities in a bend and their effects on particle impact speeds and angles.

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